ABINESH R K

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PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student (graduating June 2027) with hands-on exposure to VLSI design, RTL coding (Verilog/SystemVerilog), digital logic design, and ASIC verification methodologies. Completed a VLSI design internship covering the industry design flow from specification through RTL to verification. Delivered independent projects on pipelined CRC-based error detection for I2C and approximate parallel prefix adders, with exposure to UVM and assertion-based verification concepts. Proficient in C, C++, Python, and Linux; seeking a Design Verification Internship to contribute to RTL design and functional verification.

CORE COMPETENCIES

Programming & Tools: C, C++, Python, MATLAB, Vivado, Proteus, ModelSim, Git/GitHub

Embedded Platforms: Arduino, Raspberry Pi, 8051/ARM microcontroller

EDUCATION

B.E. Electronics & Communication Engineering — **2023 – 2027 (Final Year)**
Kongunadu College of Engineering & Technology

Higher Secondary Certificate (Class XII) — 71.2% — **2023**
Thamarai International School, Kumbakonam

Secondary School Certificate (Class X) — 83.4% — **2021**
Thamarai International School, Kumbakonam

PROFESSIONAL EXPERIENCE

VLSI Design Intern — **Jun 2025**
SD PRO Solution Pvt. Ltd., Trichy

- Completed an intensive VLSI internship covering digital logic design, semiconductor device concepts, and the industry ASIC design flow (specification → RTL → verification).
- Gained hands-on exposure to combinational and sequential circuit design; coded RTL modules in Verilog and simulated using standard HDL workflows.
- Studied HDL-based simulation and verification methodologies; executed testbenches and analysed waveforms for functional debugging.
- Assisted in understanding physical design and synthesis concepts, strengthening knowledge of STA and timing closure.
- Collaborated with senior engineers to understand cross-functional VLSI workflows spanning design, DFT, and physical design.

ACADEMIC PROJECTS

Efficient Error Detection for I2C Protocol using Pipelined CRC — **Oct 2025**
VLSI / RTL Design — Verilog, HDL Simulation

- Designed a pipelined Cyclic Redundancy Check (CRC) module in Verilog to enhance data integrity in I2C communication.
- Implemented parallel-bit pipeline stages, reducing CRC computation latency and increasing throughput for real-time embedded systems.
- Developed RTL testbenches to simulate and verify functional correctness of the error-detection logic across varied data patterns.

Approximate Parallel Prefix Adders — **Aug 2025**
VLSI / Digital Design — Verilog, HDL Simulation

- Implemented and compared approximate parallel prefix adder architectures (Kogge-Stone, Brent-Kung variants) in Verilog targeting DSP and AI-accelerator applications.
- Analysed trade-offs between area, power, delay, and acceptable error rate — core skills for low-power ASIC design.
- Performed RTL simulation and waveform analysis to validate functional behaviour and quantify error metrics.
- Studied synthesis-level area and timing estimation, building foundational knowledge of STA and synthesis flows.

CERTIFICATIONS

- VLSI Design Fundamentals — SD PRO Solution Pvt. Ltd.
- Embedded Systems with AI — Galvin Tech
- Steganography using PBTL Technique — Sriram Engineering, THINKFINTY-2K25

ADDITIONAL INFORMATION

Languages: Tamil (Native), English (Professional) | Interests: Reading technical content, storytelling and visual design